TurboQuant v1.2.0

Work Log

2026-03-31 21:00 ~ 04-01 05:47

Total 8h 47m

llama.cpp KV Cache Compression

AmesianX

Session Overview

- 21:00 ~ 21:30 — Status review + work planning
- 21:30 ~ 22:30 — v2 Kronecker bug analysis
- 22:30 ~ 00:30 — V cross-head WHT implementation
- 00:30 ~ 02:00 — Scientist name test (quality verification)
- 02:00 ~ 02:30 — Paper verification + head_dim limitation confirmed
- 02:30 ~ 03:00 — head_dim=64 solution
- 03:00 ~ 04:00 — Auto head_dim Mapping implementation
- 04:00 ~ 05:47 — Review, benchmark, deployment, community

v2 Kronecker Bug Analysis Debug

- Deep comparison of v1 (serial 512-WHT) vs v2 (Kronecker $H_8 \otimes H_{64}$) in `set-rows.cu`
- Full verification of sign patterns, butterfly direction, `__syncthreads__` placement, warp shuffle mask
- Standalone CUDA test written — `test-xhead-v1-v2.cu`

Conclusion: v1 == v2 (max error 1e-6) — Algorithm itself is correct. PPL difference was a build/integration issue.

V Cross-Head WHT Implementation

New Feature

Core Math

$$V_{real,h}[j] = (1/8)\ \Sigma_g\ H_8[h,g] \times \text{dequant}(V_{block,g}[j])$$

Modified Files

FATTN-VEC.CUH

Integrated 8-head load + H_8 inverse into V accumulation loop

Applied head-specific signs to V IWHT

FATTN.CU

Added V=_3 FATTN_VEC_CASE dispatch (8 combinations)

Added V TBQ type check to VEC routing

CMAKELISTS.TXT

Registered 8 new template instances

RESULT

PPL improved by 20%

3315 → 2649 (vs per-head V)

Scientist Name Test Key Finding

Token-level quality verification via Korean transliteration accuracy of German physicist name "Wolfgang Pauli"

QUANTIZATION	PPL	OUTPUT	VERDICT
F16 / F16	4008	"Bolpeugang Paulli" (correct)	PASS
tbqp3_3 / f16	**2195**	"2."	FAIL
tbq4_0 / f16	3421	"Bolpeugang Pauri" (1 syllable off)	1 syllable diff
tbq3_0 / f16	3089	"Paibeuraseu" (garbage)	FAIL

PPL Trap Discovered: WHT improves PPL via attention smoothing, but destroys peak precision — PPL and generation quality are inversely correlated

head_dim Limitation Confirmed

Analysis

Paper scope: TurboQuant paper only validated head_dim=128 (Gemma, Mistral, Llama-3.1-8B)
head_dim=64 fails due to insufficient CLT (Central Limit Theorem) convergence

Test Results by head_dim

64	128	256
Generation broken	Scientist names correct	Scientist names correct
NOT OK	PASS	PASS

Conclusion: TurboQuant WHT is only practical at **head_dim ≥ 128**

head_dim=64 Fix + Auto Mapping Resolved

HEAD_DIM=64 SOLUTION

- K = `q8_0` (standard 8-bit)
- V = `tbq3_2` (WHT 3-bit)
- "Wolfgang Pauli" correctly output in Korean
- 2.7x compression ratio maintained

AUTO HEAD_DIM MAPPING

- User input: `tbq3` (shorthand)
- Auto-detect head_dim from GGUF metadata
- 256 → `_0` / 128 → `_1` / 64 → K= `q8_0` +V= `_2`
- Input validation + fallback + warning box

Modified Files

- `arg.cpp` — shorthand mapping
- `common.cpp` — head_dim detection + auto selection

GPT OSS 20B/120B, Qwen3.5-27B/122B — all models passed testing

Review / Benchmark / Deployment Complete

CODE REVIEW + BUG FIXES

- Full code review using Agent (4 files)
- V wrong-suffix fallback: `q8_0` → `f16` fix
- TBQP V downgrade logic cleanup
- Rebuild + verification passed

BENCHMARK COLLECTION

- GPT OSS 20B/120B: F16 vs auto-mapped
- Qwen3.5-122B: F16 vs tbqp3/tbq3
- KV size, compression ratio, speed, PPL comparison

Deployment

- Patch files generated (full + 13 per-file), modified_files updated (68 files)
- main push → **v1.2.0 tag** → GitHub release created → workflow triggered
- Updates shared on llama.cpp discussion #21155, #20969

Key Achievements Summary

6	8h 47m	v1.2.0
Key Achievements	Work Duration	Release Deployed

- **V cross-head WHT implemented** — 20% PPL improvement vs per-head V
- **PPL trap discovered** — head_dim=64: PPL ↓ but generation quality ↓ (inverse correlation)
- **Auto head_dim mapping** — Just input `tbq3` for automatic head_dim selection
- **head_dim=64 resolved** — K=q8_0 fallback + V=tbq3, Pauli test passed
- **Input validation hardened** — Any combination handled safely without crashes
- **v1.2.0 released + community update**

Thank You

TurboQuant v1.2.0

2026-04-01